Table 1: GCI–Richardson extrapolation vs. Roark: deflection and rotation.

Job	Load type	QoI	ϕ_1	GCI_{12} (%)	GCI_{23} (%)	$\mathrm{GCI}_{23}/\mathrm{GCI}_{12}$	p	r^p	$\Delta(\phi_1, \mathrm{Roark})$ (%)	ϕ_{rich}
0	End	u_y (mm)	-3.041	—	—	—	—	—	-0	-3.041
0	End	θ_z (deg)	-0.1307	—	—	—	—	—	-0	-0.1307
1	Mid-span	u_y (mm)	-0.9504	—	—	—	—	—	-0	-0.9504
1	Mid-span	θ_z (deg)	-0.0327	—	—	—	—	—	-0	-0.0327
2	Quarter-span	u_y (mm)	-0.2614	—	—	—	—	—	-0	-0.2614
2	Quarter-span	θ_z (deg)	-0.0082	—	—	—	—	—	-0	-0.0082
5	UDL	u_y (mm)	-2.281	0.004	0.017	4	2	4	0	-2.281
5	UDL	θ_z (deg)	-0.0871	0.006	0.025	4	2	4	0	-0.0871
6	Triangular	u_y (mm)	-1.673	0.003	0.011	4	2	4	0	-1.673
6	Triangular	θ_z (deg)	-0.0653	0.005	0.02	4	2	4	0	-0.0653
7	Parabolic	u_y (mm)	-1.318	0.002	0.01	4	2	4	0	-1.318
7	Parabolic	θ_z (deg)	-0.0523	0.005	0.019	4	2	4	0	-0.0523

Note. Grids: $n_1 = 100$, $n_2 = 50$, $n_3 = 25$ elements; refinement ratio $r = 2$; safety factor $F_s = 1.25$

(three-grid): we use the standard three-grid GCI formula, and the factor 1.25 is the one recommended for that case. ϕ_1 = fine-grid solution; p = apparent order ($p_{\mathrm{th}} = 2$). $\Delta(\phi_1, \mathrm{Roark}) = 100(\phi_1 - \mathrm{Roark})/\mathrm{Roark}$; ϕ_{rich} = Richardson extrapolate. $\mathrm{GCI}_{23}/\mathrm{GCI}_{12} \approx r^p$ in asymptotic range.